

Duality

Design and Analysis of Algorithms

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Two Related Problems

Given N intervals $[a_i, b_i]$:

- **Activity selection:** Select a MAXIMUM number of intervals, no two of which can overlap.
- **Disconnection:** Make the MINIMUM number of cuts on the number line so that every interval is severed.

Two Related Problems

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- **Disconnection:** Make the MINIMUM number of cuts on the number line so that every interval is severed.

If A is any feasible solution to the first problem and B is any feasible solution to the second problem, then it's easy to show that

$$\# \text{ intervals in } A \leq \# \text{ cuts in } B.$$

(since every interval in solution A needs a different cut)

Many Useful Consequences...

E.g., Certifying Optimality

- **Activity selection:** Select a MAXIMUM number of intervals, no two of which can overlap.
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Let A be any feasible solution to the first problem and B be any feasible solution to the second problem. If

$$\# \text{ intervals in A} = \# \text{ cuts in B.}$$

then both solutions are optimal!

Many Useful Consequences...

E.g., Certifying Optimality

- **Activity selection:** Select a MAXIMUM number of intervals, no two of which can overlap.
- **Disconnection:** Make the MINIMUM number of cuts on the number line so that every interval is severed.

Let A be an optimal solution to the first problem and B be an optimal solution to the second problem. These two problems satisfy:

$$\# \text{ intervals in A} = \# \text{ cuts in B.}$$

This is sometimes known as “strong” duality, whereas the $\leq$ relationship only is known as “weak” duality.

Bounding Distance to Optimality

- **Activity selection:** Select a MAXIMUM number of intervals, no two of which can overlap.
- **Disconnection:** Make the MINIMUM number of cuts on the number line so that every interval is severed.

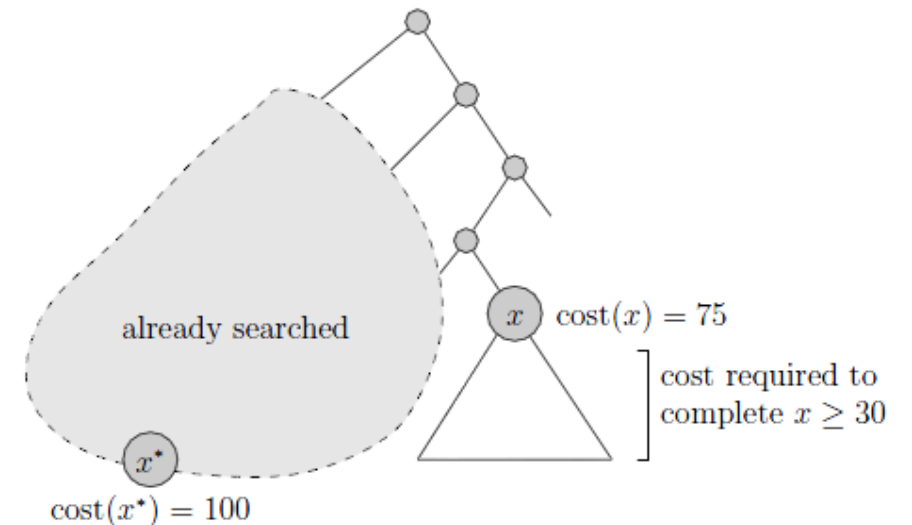
Let A be any feasible solution to the first problem and B be any feasible solution to the second problem. Then both solutions are at most

$$(\# \text{ cuts in B}) - (\# \text{ intervals in A})$$

away from being optimal.

Bounding Distance to Optimality: Applications

- **Iterative Refinement:** Gives us a sense of when we might want to stop refining.
- **“Primal-Dual” Algorithms:** Use relationships between primal problem and dual to build or refine solutions to both simultaneously.
- **Branch and Bound:** Key ingredient was finding a good lower bound on cost of completing a partial solution.
- **Approximation Algorithms:** getting a solution provably close to optimal without knowing the value of an optimal solution (this module’s enrichment lecture).



Algorithmic Advantages

- Given an array $A[1..n]$, what is the length of its longest increasing subsequence (A).

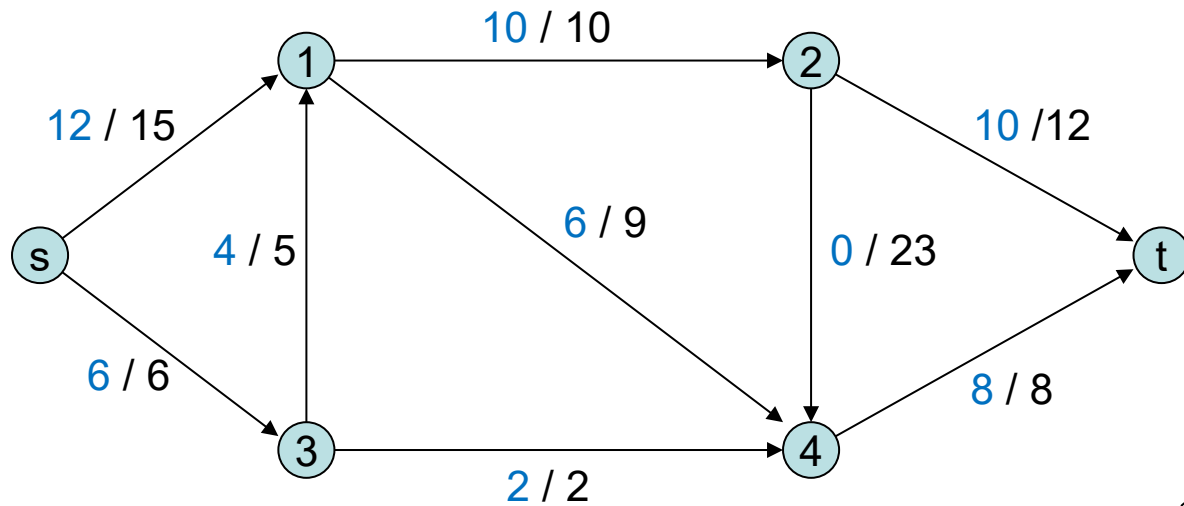
Example: 14 8 12 9 7 4 11 15 6 20 -3

- Dual problem: partition an array into a minimum number of decreasing subsequences (B).

Example: **14** 8 **12** 9 7 4 11 15 6 20 -3

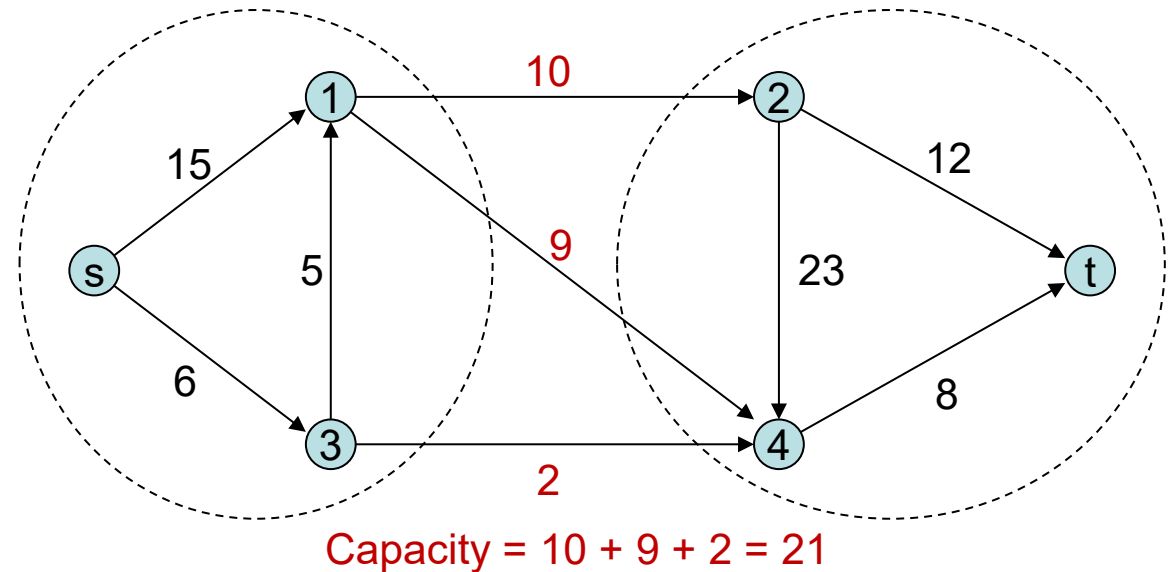
- Easy to see $A \leq B$, since every element in a solution to A must come from a different decreasing subsequence.
- An $O(n \log n)$ greedy “sweep line” algorithm solves B, and with care, an optimal solution of A can also be recovered.

Prominent Classes of Problems Satisfying Duality Relationships

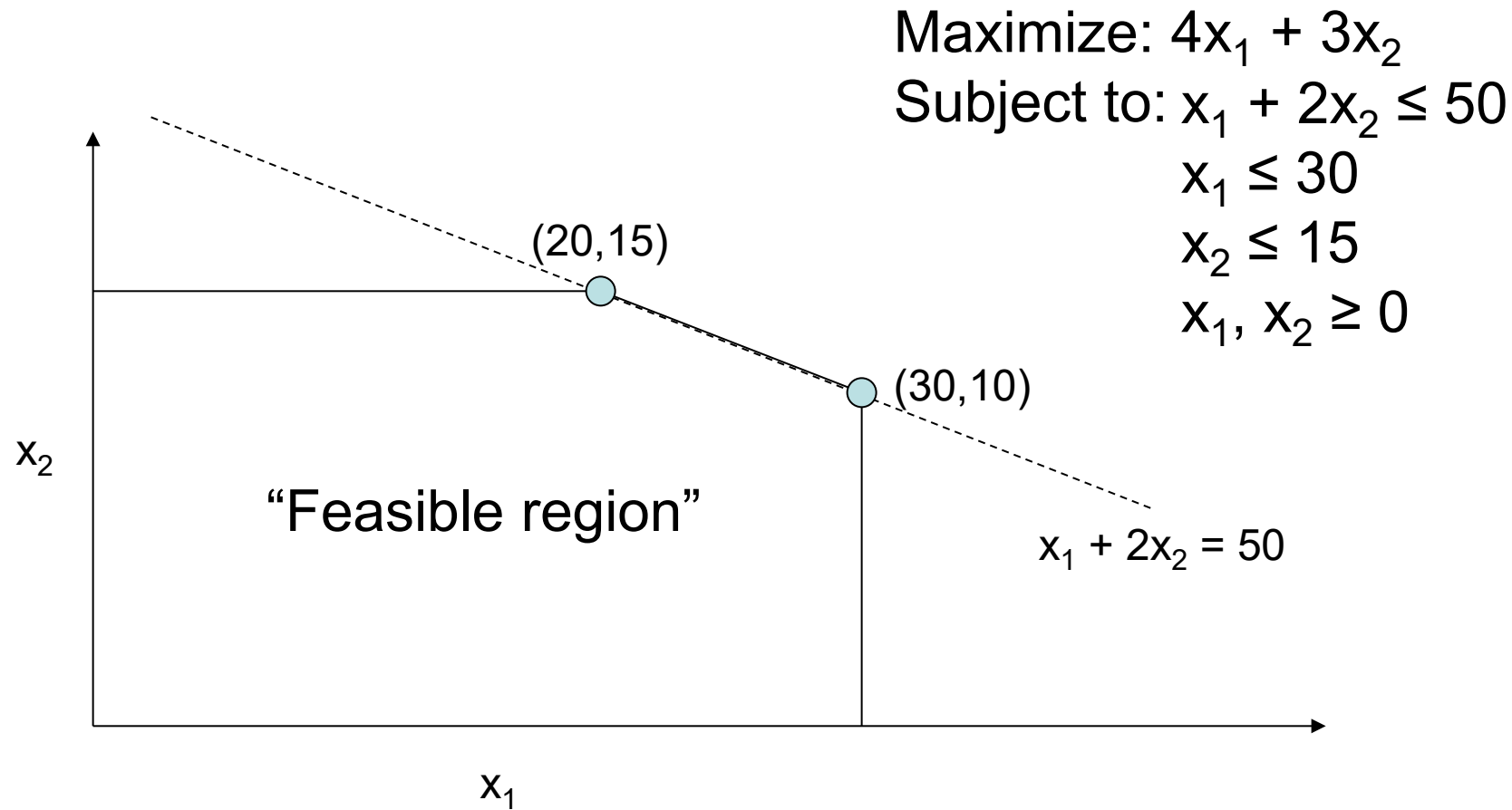


Maximum s – t Flow

Minimum s – t Cut



Prominent Classes of Problems Satisfying Duality Relationships



Linear Programming